Sumit Singh Bagdwal

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Education

Graphic Era Hill University — Bhimtal, Uttarakhand
B.Tech in Computer Science and Engineering | CGPA: 8.26/10 — *2024 – Expected 2028*

Technical Skills

Languages: C++, JavaScript (ES6+), Python, SQL
Backend & Systems: Node.js, Express.js, FastAPI, REST APIs, JWT, RBAC, WebSockets, Socket.IO
Databases & Cloud: MySQL, MongoDB, Redis, AWS (EC2, S3, IAM), Docker, Nginx, Railway, Vercel
Frontend & Tools: React.js, HTML/CSS, Tailwind CSS, Git, GitHub, Postman, GitHub Copilot, Claude Code
Foundations: Data Structures & Algorithms, OOP, DBMS, Agile/Scrum, Prompt Engineering

Projects

IBVAP | *Node.js, Python, FastAPI, YOLO11, ByteTrack, AWS* — GitHub

- Built a **Smart India Hackathon 2026 shortlisted** video analytics prototype that detects and tracks objects in CCTV streams; combined spatial/temporal rules with **per-track alert deduplication** and severity escalation.
- Separated **FastAPI edge inference** from a Node.js/Express event backend; connected a private inference node to a Dockerized **AWS EC2** stack (Nginx, MySQL, Redis) over Tailscale for secure streaming and updates.
- Added **SHA-256/Ed25519 evidence integrity**, RBAC, realtime alerts, and camera reconnection handling; documented **407 passing AI tests and 41 frontend tests** in local validation.

Pratyaya | *Node.js, Express, React, MongoDB Atlas, Mongoose* — GitHub

- Building a **backend-first payment simulation** using Node.js/Express, React, and **MongoDB Atlas**, with merchant-scoped payment workflows and role-based sandbox exploration.
- Designed **idempotent REST APIs**, a **double-entry ledger**, and a transactional outbox using MongoDB ACID transactions to model duplicate-request safety and auditable financial records.
- Architected a **payment reliability lab** with independent mock providers, failure injection, durable background jobs, and reconciliation for uncertain outcomes such as timeouts and duplicate webhooks.

Leadership & Achievements

Lead, AWS Student Builder Group | *Graphic Era Hill University, Bhimtal* — 2026 – Present

- Lead student workshops and hands-on learning initiatives in backend development, cloud, and problem solving; helped organize the **INNOVATE 1.0** campus ideathon.

- **Competitive achievements:** Smart India Hackathon 2026 shortlist; Hack the Winter 2025 national finalist; **350+ LeetCode problems** solved.
- **Learning & service:** NPTEL Data Structures and Design; AWS Cloud Essentials and Practitioner Essentials training; NCC B Certificate.